Calcium Channel Blocker Overdose

Support airway, breathing and circulation

- Maintain Sp02 94-98%
- ETCO2 monitoring
- Gain IV access
- **12-Lead ECG**
- **Continuous ECG monitoring**

Consider ALS Back-Up

Signs of Shock

Administer IV Fluid

- **Adults:** 500mL bolus, repeat as required, maximum 2000mL.
- **Pediatric:** 20 mL/kg, repeat as required, maximum 80mL/kg.

EMT

Symptomatic Bradycardia

Administer Atropine

- **Adults: 1mg IV/IO**
  Every 3-5 minutes (max 3mg)
- **Pediatric: 0.02mg IV/IO**
  (maximum 0.5mg single dose) every 3-5 minutes (maximum 1mg total)

EMT-A

AP

Symptomatic Bradycardia Refractory to atropine
or
Shock refractory to fluid administration

Administer IV Adrenaline

	Adult	Pediatric
Push Dose	**0.1mg** IV/IO	**0.001mg/kg** IV/IO
Infusion	**0.05-0.3mcg**/kg/min	**0.05-0.3mcg**/kg/min

Symptomatic Bradycardia or Shock Refractory to IV Adrenaline

Administer Glucagon

Adults: 1mg IM

- Increase dose to 5mg IM if additional glucagon available.

Pediatric:

- **<25kg: 0.5mg IM**
- **>25kg: 1mg IM**

EMT

Ongoing Symptomatic Bradycardia

Consider Transcutaneous Pacing

Patient Care Goals

1. Rapidly identify and stabilize patients with suspected calcium channel blocker (CCB) overdose to prevent progression to severe hemodynamic instability, shock, or cardiac arrest.
2. Provide supportive care to maintain airway, breathing, circulation.
3. Ensure timely transport to a facility capable of advanced interventions.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with signs and symptoms suggestive of CCB overdose, including:
  - Ingestion of CCBs (e.g., verapamil, diltiazem, amlodipine, nifedipine) in amounts exceeding therapeutic doses.
  - Hemodynamic instability: bradycardia, hypotension, heart block, or shock.
  - Systemic symptoms: altered mental status, hyperglycemia, seizures, nausea/vomiting, or metabolic acidosis.
  - History of intentional overdose or accidental ingestion (e.g., in children).

Exclusion Criteria

- None Noted.

Key Patient Assessment Points

1. **History and Exposure**
   - Obtain details: Type of CCB (immediate vs. sustained-release), time and quantity ingested, co-ingestants (e.g., beta-blockers, digoxin).
2. **Vital Signs Assessment**
   - Continuous cardiac monitoring,
   - 12-lead ECG (to identify bradycardia, or arrhythmias).
   - Blood glucose (hyperglycemia is common in CCB toxicity).
3. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - Patients presenting with moderate bradycardia or arrhythmias following Calcium Channel Blocker overdose should be considered red priority.

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Patient Management & Interventions

1. **Primary & Secondary Survey**
   - Perform as per Universal Care CPG & General Toxicology Management CPG.
2. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG
3. **IV Access**
   - Establish IV access
   - Consider IO access if IV access is not readily accessible
4. **Fluid resuscitation**
   - Indications:
     - Signs of shock
   - Dose:
     - Adults: 500mL bolus, repeat as required, maximum 2000mL.
     - Pediatric: 20 mL/kg, repeat as required, maximum 80mL/kg.
5. **Atropine**
   - Indications:
     - Symptomatic Bradycardia
   - Administer Atropine:
     - Adults: 0.5-1mg IV/IO every 3-5 min (max 3 mg)
     - Pediatric: 0.02mg/kg, (0.5 mg maximum single dose), every 3-5 minutes (maximum 1 mg).
6. **IV Adrenaline**

- Indications:
  - The patient remains hypotensive after adequate fluid resuscitation.
- Adrenaline infusion via syringe driver is preferred when available.
- Adrenaline Infusion:
  - Dose: Infusion: 0.05–0.3 mcg/kg/minute
- Push Dose Adrenaline:
  - Adult Dose: 0.1mg IV/IO
  - Pediatric: 0.01mg/kg IV/IO

7. **Glucagon**
   - Indications:
     - The patient remains hypotensive after IV adrenaline.
   - Dose:
     - If IV Glucagon is available:
       - Adult: Glucagon 5 mg IVP, then:
         - 1 mg IV/IO, every 5 minutes IVP (may require 5–15 mg to see effect)
       - Pediatric: (25–40 kg): Glucagon 1 mg IV, repeat every 5 minutes as required.
       - Pediatric: (less than 25 kg): Glucagon 0.5 mg IV, every 5 minutes as required.
     - **If IV Glucagon is not available, Administer IM Glucagon**:
       - Adults: 1mg IM.
         - Administer up to 5mg IM (at different injection sites) if additional IM glucagon is available.
         - Repeat every 10-15 minutes if required.
       - Pediatrics (<20kg): 0.5mg IM.
       - Pediatrics (>20kg): 1mg IM.
         - Repeat every 10-15 minutes if required.
8. **Transcutaneous Pacing**
   - Implement transcutaneous pacing if refractory to pharmacologic interventions.
9. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.

Patient Safety Considerations

- Time-Sensitive Transport
  - Rapid transport for symptomatic patients

Clinical Insights & Additional Guidance

- Differentiation by CCB Type:
- Verapamil/diltiazem more cardiotoxic (bradycardia); dihydropyridines (e.g., amlodipine) cause vasodilation and reflex tachycardia initially.

Documentation

- Identification: Patient's full name, age, and gender.
- Mechanism / Medical Complaint: CCB type, dose, time of ingestion, co-ingestants.
- Injuries / Information: Symptoms progression, differentials ruled out (e.g., beta-blocker).
- Signs: Vital signs, ECG findings, glucose levels, severity.
- Treatment:
  - Interventions (e.g., glucagon, vasopressors).
  - IV fluids and access.
- Allergies: Known allergies.
- Medications: Chronic meds (e.g., cardiac drugs).
- Background: Relevant history (e.g., hypertension, prior overdoses).
- Other: None Noted.

Key Performance Indicators

1. Investigate and document substance type, route, amount, and exposure/ingestion time.

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